December Research: Baseline vs. Agent

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Objective

For the December deliverable, my partner and I will conduct a quantitative benchmark evaluation comparing the Baseline RAG system with the Agent.

The goal is to determine whether the Agent provides measurable improvements over the Baseline when both systems are tested on the same ground-truth benchmark.

Evaluation Plan

Both systems will receive the same requirements and their results will be compared against the expected NASA SWE clause and compliance verdict.

The main metrics I plan to evaluate are:

- **Valid Output Rate:** Whether the system produces the required structured output.
- **Clause Retrieval:** Whether the correct NASA SWE clause was retrieved.
- **Clause Citation:** Whether the correct clause was actually used/cited in the final result.
- **Verdict Accuracy:** Whether the predicted compliance verdict matches the expected verdict.
- **Grounded Verdict Accuracy:** Whether the verdict is correct and supported by the correct NASA clause.

- **Precision and Recall:** How accurately the system identifies real compliance gaps without creating too many false positives or missing actual gaps.

Planned Process

The evaluation will follow this general process:

Benchmark → Run Baseline → Run Agent → Save Results → Calculate Metrics → Compare Performance

The repository already tracks several useful evaluation fields, including expected_clause_retrieved, expected_clause_cited, verdict_matches, and grounded_verdict_matches. Because of this, we can build on the existing evaluation work instead of creating everything from scratch.

Division of Work

Shahreen: Running the Baseline and Agent and collecting the benchmark results

Nam: Calculating the metrics, comparing the results, and analyzing errors.

We can then combine both parts for the final evaluation.

Current Research / Next Steps

From reviewing the current project, I noticed that the existing saved Baseline and Agent results should not yet be directly compared because they were not run on the exact same number of benchmark examples.

Our next step will therefore be to run both systems on the same benchmark under the same conditions. After that, we can create a comparison table showing which system performs better and analyze situations where the Baseline and Agent produce different results.

The final goal is to determine, using measurable results, whether adding the Agent improves the compliance-checking system compared with the original Baseline.

Sources:

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